Reg No.

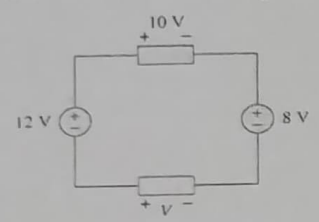
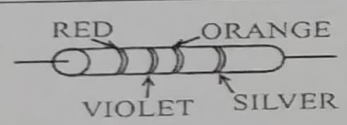
SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT TEST – I

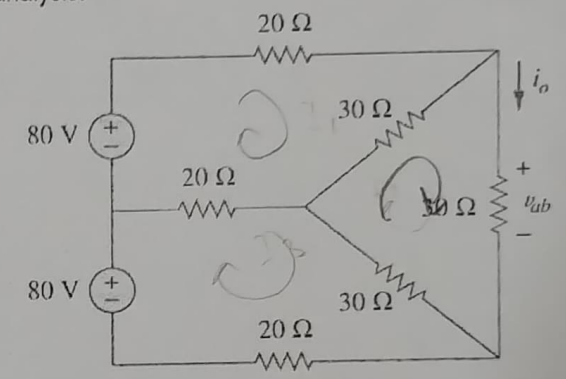
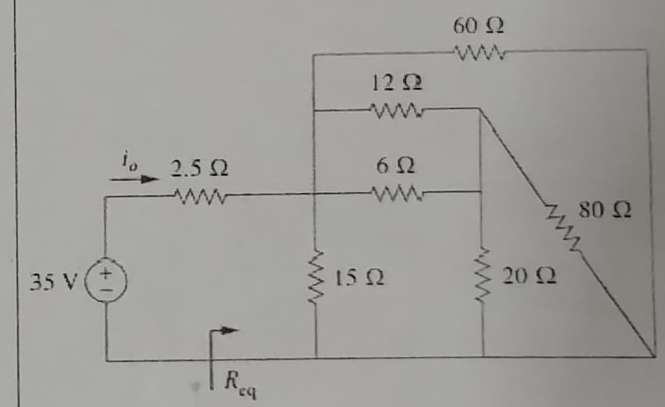
Course: B.E-ECE-DS
Subject: Circuit Theory& Network Analysis
Sub.code:S14BLH24
Batch: 2024-2028

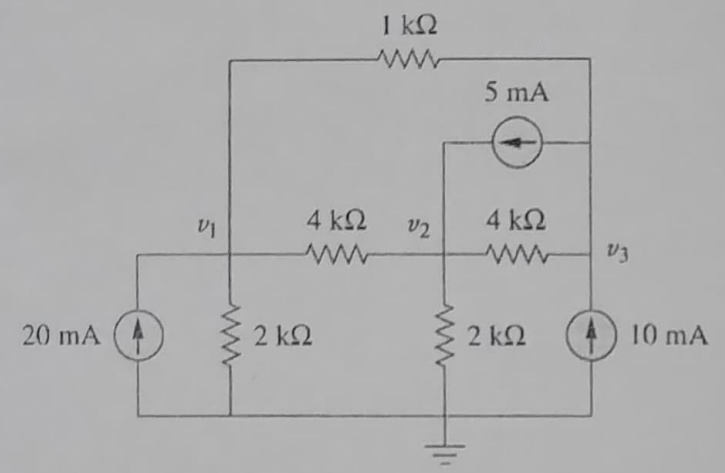
Max. Marks: 50
Time: 2 Hours
Sem: II
Date: 23-2-2026

Part-A AnswerALL the questions (7×2=14)

Q. No	Questions	CO
1.	Differentiate Electric Power and Electrical Energy	1
2.	Find the value of V in the given diagram 	1
3.	Why the non-diagonal element is negative in mesh analysis.	1
4.	 Find the value of resistance of the above diagram.	1
5.	In a+jb rectangular format what does J represents.	2
6.	Define average value.	2
7.	Write the properties of vector list out them.	2

Part-B Answer ALL the questions (3×12=36)

Q.No	Questions
8.	Find the currents in the mesh and the voltage in the 30 Ω resistor by mesh analysis. 
	(OR)
9.	Obtain the equivalent resistance and the current in 2.5 Ω resistor. 

10.	Find the voltage in the major node and the power dissipated in the 1 kΩ resistor by nodal analysis method 	1
	(OR)	
10.	Derive the expression for current and voltage and draw the phasor diagram, waveform for pure inductor and discuss its behavior when its is subjected to sinusoidal excitation.	2

11.	Derive the average value and RMS value for input waveform of rectifier.	2
	(OR)	
12.	A coil connected to a single phase 230-volt, 50 Hz AC supply dissipates 1.5kW of power, when it takes 10 amps of current from the supply. Find the resistance and reactance of the coil. Also determine phase angle and draw the phasor diagram.	2